UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 **(5 marks)**

a) Describe ONE (1) major difference between a tightly coupled multiprocessor system and a loosely coupled multiprocessor system.

b) State the THREE (3) types of cluster architecture.

QUESTION A2 **(5 marks)**

Convert 100.5 to IEEE 754 single-precision format. Give the answer in hexadecimal and show the sign, exponent and mantissa fields.

QUESTION A3 **(5 marks)**

Question 1 is based on the following Figure 1.

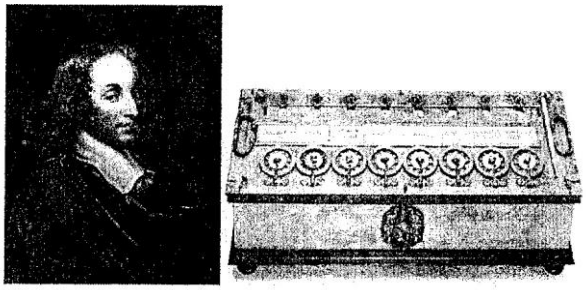


Figure 1: Inventor 'Y' and his calculating machine.

Inventor 'Y' was France's most celebrated mathematician and physicist. In 1642, inventor 'Y' invented and build the first working calculating machine as a means of helping his father to perform tedious tax accounting.

a) Who is inventor 'Y'?

b) Name another calculating machine that was invented in the same generation as inventor Y's calculating machine.

c) Which programming language is named after inventor 'Y'?

d) State TWO (2) mathematical operationsthat can be performed byinventor Y's calculating machine.

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QUESTION A4 **(5 marks)**

Given the following Figure 2.

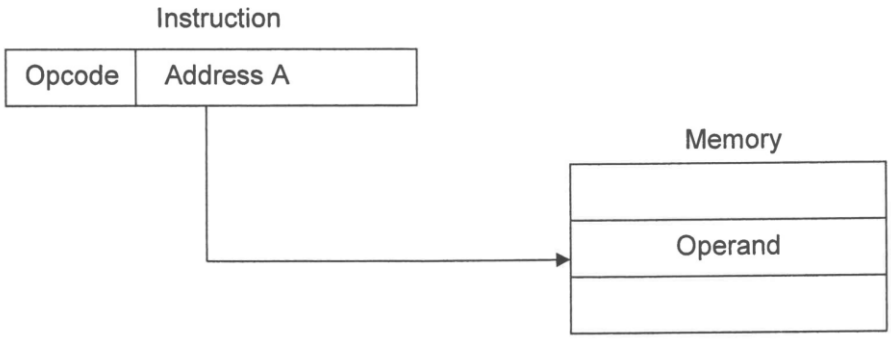


Figure 2

a) Label the addressing technique illustrated in Figure 2.

b) Explain how the addressing technique in 9(a) is used.

c) List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in 9(a).

QUESTION A5 **(10 marks)**

a) Differentiate the involvement of CPU in the interrupt-driven I/O technique and Direct Manipulation Access

b) Define interrupt.

c) Discuss what happens during a context switch.

d) Elaborate on the benefit of a context switch.

QUESTION A6 **(10 marks)**

Answer both parts. Show your complete workflow.

a) Convert 159910 to hexadecimal.

b) Convert FD916 to decimal.

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QUESTION A7 **(10 marks)**

a) Discuss THREE (3) types of interfaces to external devices.

b) Classify the following input/output devices to their respective 1/0 module:

i. Network interface card Scanner Touch Screen

i. Register

QUESTION A8 **(10 marks)**

Describe the specific function of each registers in the CPU.

QUESTION A9 **(10 marks)**

a) List FOUR (4) types of access method for storage device with ONE (1) example for each of them.

b) Calculate the total of memory location using 11 address bit.

c) Calculate the address bits required for 2MB of RAM.

QUESTION A10 **(10 marks)**

a) Discuss the function of each Central Processing Unit (CPU) component'

b) Elaborate on what will happen in the machine cycle if the bus system is faulty

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PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 **(20 marks)**

Question 1 is based on Figure 2 and Table 1.

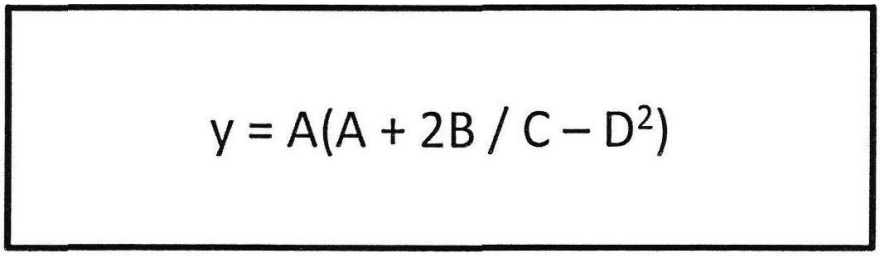


Figure 2

Write the appropriate programs using correct assembly language notations and comments to compute the equation represented in Figure 1 using zero-, one-, and three-address instructions. The instructions available for use are provided in Table 1. Register A, B, C, and D cannot be modified, and registers x, y, and z can be used as temporary registers. However, the final answer must be stored in Register y.

Table 1: Machine Instruction

Zero-address	One-address	Three-address
PUSHp	LOADp	ADD (p,q,r)
POPp	STORp	SUB (p,q,r)
ADD	ADDp	MUL (p.q.r)
SUB	SUBp	DIV(p,q,r)
MUL	MULp	
DIV	DIVp	